

Laboratory work 2 report
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CSS4007-ENG-6

I used the GPT-2 small model via Google Colab. The model contains 124 439 808 parameters, 50 257 tokens, 12 layers and 12 attention heads.

```
torch 2.11.0+cpu | transformers 5.16.1
parameters      : 124,439,808
vocabulary      : 50,257 tokens
layers x heads  : 12 x 12
embedding table : (50257, 768) (rows = tokens, columns = numbers per token; 768 is the GPT-2 small figure from slide 10)
```

At Part 0 I checked the tokenization. restore crashed on 2 tokens, bank 1, bank on 5, and бөлімшеңізде on 19 tokens. This shows that one word can consist of several tokens.

```
'restore': 2 tokens
ids      : [2118, 382]
pieces: ['rest', 'ore']
'bank': 1 tokens
ids      : [17796]
pieces: ['bank']
'банк': 5 tokens
ids      : [140, 109, 16142, 22177, 31583]
pieces: ['ð', '±', 'ðº', 'ð¼', 'ðº']
'бөлімшеңізде': 19 tokens
ids      : [140, 109, 143, 102, 30143, 141, 244, 43108, 141, 230, 16843, 142, 96, 141, 244, 140, 115, 43666, 16843]
pieces: ['ð', '±', 'ó', 'ø', 'ð»', 'ñ', 'κ', 'ð¼', 'ñ', 'ī', 'ðµ', 'ò', 'ε', 'ñ', 'κ', 'ð', '·', 'ð', 'ðµ']
```

In Part 1, I checked the distribution of the probability of the next token. For the prompt about the capital of France, the most likely was Ast 22.87%, and Paris 22.08%. Ten most likely tokens together received 64% of probability. I also checked that Paris and Paris are different tokens and have highly different probabilities.

```
prompt: 'The capital of Kazakhstan is Astana. The capital of France is'   T=1.0
logits shape: (50257,)   probabilities sum to 1.0000
22.87% id 8304 'Ast'
22.08% id 6342 'Paris'
4.48% id 262 'the'
3.15% id 18460 'Nice'
2.56% id 4881 'France'
2.15% id 31630 'Monaco'
1.97% id 9281 'Saint'
1.95% id 45996 'Marse'
1.51% id 35193 'Lyon'
1.23% id 347 'B'
top-10 hold 64.0%; the other 50,247 tokens share 36.0%
```

When temperature changes from 0.25 to 5, the first token has not changed it has always been Ast. Its probability decreased from 53.45% to 0.04%, and entropy increased. So the temperature changes the distribution of probability, but does not change the most likely token.

```

prompt: 'Once upon a time, there was a'    T=1.0
logits shape: (50257,)    probabilities sum to 1.0000
  4.42%   id    582    ' man'
  3.31%   id   1049    ' great'
  2.20%   id    640    ' time'
  1.54%   id   1728    ' certain'
  1.34%   id   2415    ' woman'
top-5 hold 12.8%; the other 50,252 tokens share 87.2%

```

```

prompt: 'The capital of France is'    T=1.0
logits shape: (50257,)    probabilities sum to 1.0000
  8.46%   id    262    ' the'
  4.79%   id    783    ' now'
  4.62%   id    257    ' a'
  3.24%   id   4881    ' France'
  3.22%   id   6342    ' Paris'
top-5 hold 24.3%; the other 50,252 tokens share 75.7%

```

```

prompt: '2 + 2 ='    T=1.0
logits shape: (50257,)    probabilities sum to 1.0000
 10.19%   id    513    ' 3'
   9.16%   id    352    ' 1'
   8.42%   id    362    ' 2'
   7.52%   id    657    ' 0'
   5.50%   id    604    ' 4'
top-5 hold 40.8%; the other 50,252 tokens share 59.2%

```

```

P(' Paris') = 0.2208
P('Paris') = 0.000009

```

maximum possible entropy (all 50,257 tokens equally likely): 10.825 nats

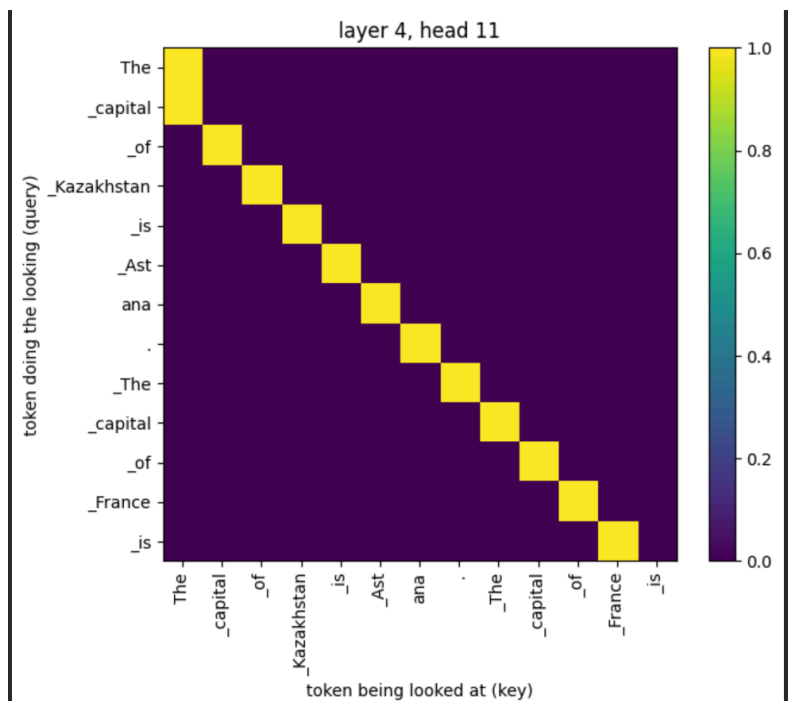
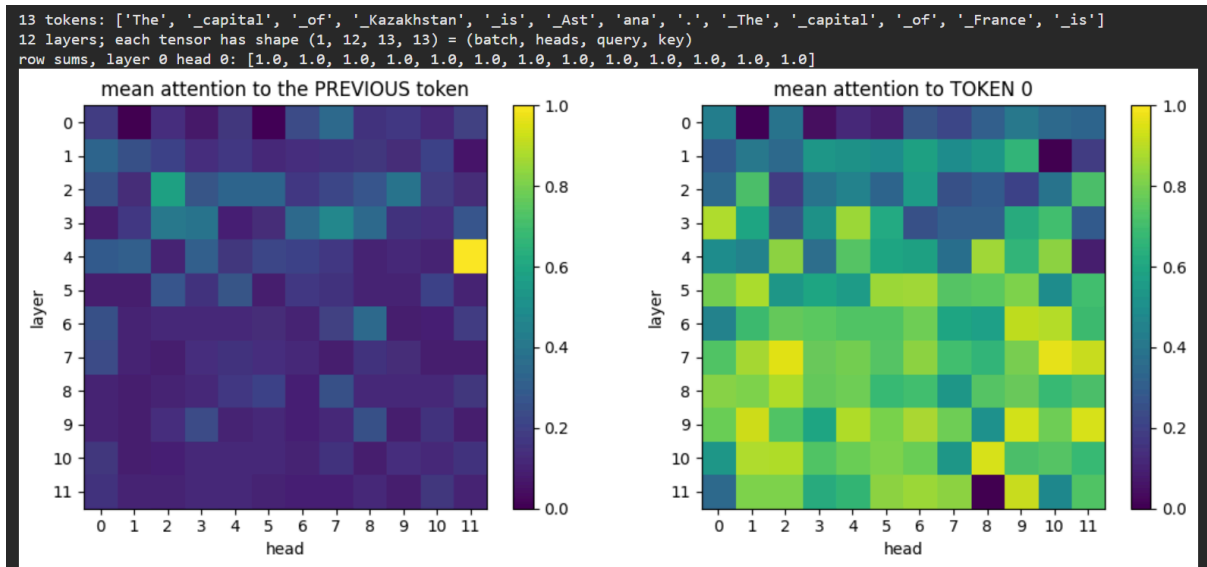
T	#1 token	p(#1)	entropy	same #1 as at T=1?
0.25	' Ast'	0.5345	0.700	True
1.0	' Ast'	0.2287	4.120	True
2.0	' Ast'	0.0123	9.114	True
5.0	' Ast'	0.0004	10.547	True

Top-p removes unlikely tokens and leaves only part of distribution. When top-p = 0.9 the number of remaining tokens depended on temperature: from 2 to 35 021.

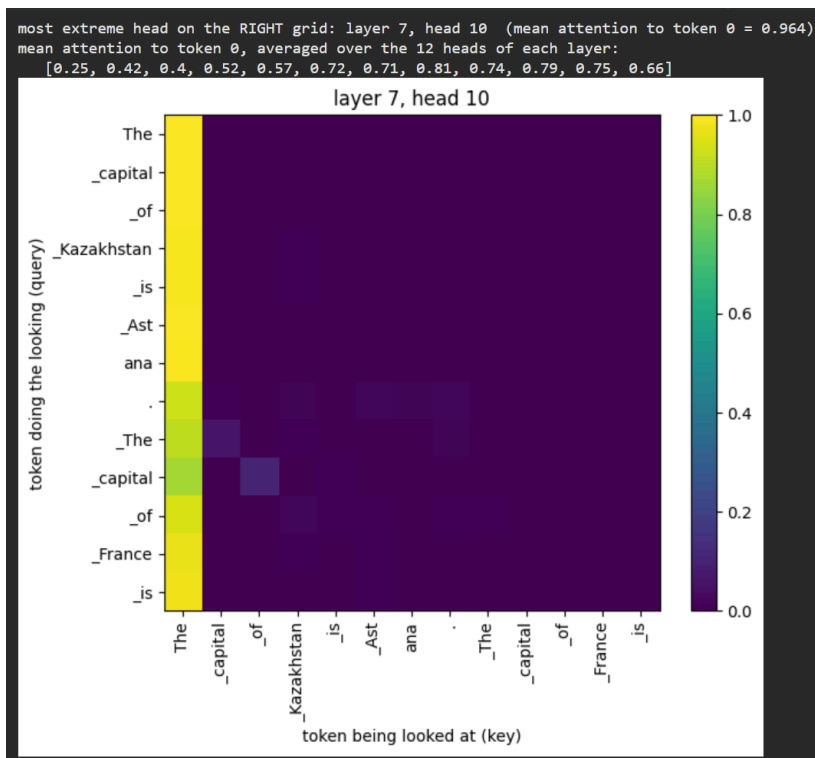
T	tokens kept by top-p=0.9	p(#1) before	p(#1) after
0.25	2	0.5345	0.5351
1.0	341	0.2287	0.2541
5.0	35,021	0.0004	0.0004

T=0.25 10 samples: [' Ast', ' Ast', ' Paris', ' Ast', ' Ast', ' Ast', ' Paris', ' Paris', ' Ast', ' Paris']
 T=1.0 10 samples: [' Siem', ' Ast', ' Paris', ' Monaco', ' Ast', ' Lie', ' Pr', ' Paris', ' Ast', ' Paris']
 T=5.0 10 samples: [' Catalyst', ' Shim', ' offline', ' alienation', ' hangs', ' Deutsche', ' Mexico', ' "[', ' Actions', ' stability']
 T=1.0 + top-p 0.9: [' Mé', ' Ast', ' Paris', ' Monaco', ' Ast', ' Saint', ' Or', ' Paris', ' Ast', ' Paris']

In Part 2 I studied attention. The most pronounced previous-token head was layer 4, head 11 1,000 attention to the previous token. On the thermal map it is seen as diagonal below the main. The rest of the heads of this layer work differently.



The most expressed head on attention to the first token was layer 7, head 10 with a value of 0.964.



The greatest average values were observed in layers 5-10. The attention-weight shows the mechanism of the model's work, but in itself is not an explanation of its answer.

In Part 3 I compared English, Russian and Kazakh texts. It turned out 0.22, 1.11 and 1.10 token on the symbol respectively. For Russian and Kazakh prompts, the model also generated incorrect continuation. This is due to the peculiarities of tokenization and limited representation of languages in GPT-2 dictionary.

```
prompt: 'The capital of Kazakhstan is' T=1.0
logits shape: (50257,) probabilities sum to 1.0000
8.98% id 262 ' the'
7.63% id 1363 ' home'
5.89% id 257 ' a'
top-3 hold 22.5%; the other 50,254 tokens share 77.5%
#1 token as raw text: 'ôthe'
greedy, 20 tokens: "The capital of Kazakhstan is the capital of the country's former Soviet Union, and the country's president, Nursultan Nazar"
```

```
prompt: 'Столица Казахстана —' T=1.0
logits shape: (50257,) probabilities sum to 1.0000
37.79% id 12466 ' ♦'
2.40% id 198 '\n'
2.34% id 220 ' '
top-3 hold 42.5%; the other 50,254 tokens share 57.5%
#1 token as raw text: 'ĠD'
greedy, 20 tokens: 'Столица Казахстана — Столица Казахста'
```

```
prompt: 'Қазақстанның астанасы —' T=1.0
logits shape: (50257,) probabilities sum to 1.0000
22.54% id 12466 ' ♦'
4.11% id 220 ' '
2.39% id 198 '\n'
top-3 hold 29.0%; the other 50,254 tokens share 71.0%
#1 token as raw text: 'ĠD'
greedy, 20 tokens: 'Қазақстанның астанасы — проссии проссии про'
```